

Annibale Cois, Curriculum Vitæ

Personal Details

Surname: Cois
First Name: Annibale
Title: Dr
Physical Address: Division of Health Systems and Public Health, Department of Global Health, Stellenbosch University
Tygerberg Campus
Francie van Zijl Drive, Tygerberg, Cape Town, South Africa

Postal Address: Private Bag X1,
Matieland, 7602,
Stellenbosch, South Africa

Email: acois@sun.ac.za
ORCID [0000-0002-7014-6510](https://orcid.org/0000-0002-7014-6510)
WoS [AAK-8603-2020](#)

Education

- 2017 University of Cape Town, School of Public Health and Family Medicine, Cape Town
Ph.D. in Public Health
- 2012 University of Cape Town, School of Public Health and Family Medicine, Cape Town
Masters Degree of Public Health (epidemiology track), with distinction.
- 2005 University of Cagliari, Faculty of Education, Cagliari, Italy
Semester course on Logic and Philosophy of Science
Final mark: 30/30, cum laude
Semester course on General Psychology
Final mark: 30/30, cum laude
- 1988 University of Cagliari, Faculty of Engineering, Cagliari, Italy
Masters Degree in Electrical Engineering, summa cum laude

Post-graduate training in the field of the evaluation of health interventions and advanced epidemiological methods. Among them, training courses on: • *Advanced Course on Structural Equation Modelling using Mplus*, Utrecht University, The Netherlands • *Using quantitative bias analysis with epidemiologic data*, Centre of Excellence in Epidemiological Modelling and Analysis, Stellenbosch, South Africa • *Structural Equation Modelling*, Istanbul University, Istanbul, Turkey • *Robust statistical methods for trial design and analysis*, Novartis & Stellenbosch University, Stellenbosch, South Africa • *Evaluation of effectiveness and evidence-based health interventions*, Azienda ASL 8, Cagliari, Italy • *Evaluation of the effectiveness of health interventions: the Cochrane Drugs and Alcohol group*, ISS, Istituto Superiore di Sanità, Rome, Italy

Current Position

- 2021 - Stellenbosch University, Department of Global Health, Division of Health Systems and Public Health
Researcher

Invited Positions

- 2022 - Greenwich University, Institute for Lifecourse Development
Visiting Researcher
- 2018 - University of Cape Town, School of Public Health & Family Medicine, Division of Epidemiology and Biostatistics
Honorary Senior Lecturer

Recent Work Experience

- 2017 - 2021 South African Medical Research Council, Burden of Disease Unit, Cape Town
Senior Scientist, Consultant
- 2017 - 2020 Health System Trust, Health Research Unit, Cape Town
Biostatistician
- 2015 - 2017 University of Cape Town, School of Public Health and Family Medicine, Division of Epidemiology and Biostatistics, Cape Town
Senior Research Officer
- 2013-2015 Health System Trust, Health Research Unit, Cape Town
Biostatistician
- 2014 -2014 University of Cape Town, School of Public Health and Family Medicine, Cape Town
Lecturer
- 2013-2013 University of the Western Cape, School of Public Health
Lecturer
- 2012-2013 Division of disability studies, Faculty of Health Sciences, University of Cape Town
Statistical consulting

Teaching

- 2022 *Survey Methods*. MPhil in Health Systems. Stellenbosch University, Department of Global Health. Course Convenor.
- 2018-2022 *Seminars in Epidemiology & Advanced Epidemiology courses* (Structural Equation Modelling, Latent Class Analysis, Quantile Regression methods, Generalise Additive Models, Data Visualisation Methods). Master of Public Health. University of Cape Town, School of Public Health and Family Medicine.
- 2017- 2019 *Basic Biostatistics Course*. Postgraduate Diploma In TB-HIV Medicine. University of Cape Town, Faculty of Health Sciences & Desmond Tutu HIV Centre.
- 2017 *Quantile regression*. Seminar. University of Cape Town, School of Public Health and Family Medicine
- 2016 *Latent Growth and Growth Mixture Models*. Seminar. University of Cape Town, School of Public Health and Family Medicine
- 2014, 2015 *Introduction to Structural Equation Modelling for epidemiologists*. Seminars. University of Cape Town, School of Public Health and Family Medicine
- 2013 *Summer School 2013, Measuring Health and Disease II*, University of the Western Cape, School of Public Health. Course co-convenor.

Other

Member of the *South African Statistical Association (SASA)*.

Member of the *International Epidemiological Association* (IEA).

Member of the editorial board of *Medicina delle Dipendenze – The Italian Journal of the Addictions*.

Certified peer reviewer: (Web of Science reviewer profile: [AAK-8603-2020](#)).

Member from 2006 to 2010 of the Regional Board of the SITD (*Italian Society of the Addictions*).

Publications

2023

- 1) Roomaney RA, van Wyk B, Cois A, Pillay van-Wyk V. Multimorbidity patterns in South Africa: A latent class analysis. *Front Public Health*. 2023;10:1082587.

2022

- 2) Cois A, Abdelatif N, Steyn N, Turawa EB, Awotiwon OF, Roomaney RA, Neethling I, Joubert J, Pacella R, Bradshaw D, Pillay-van Wyk V. Estimating the burden of disease attributable to a diet low in fruit and vegetables in South Africa 1 in 2000, 2006 and 2012. *South African Medical Journal*. 2022;617–26.
- 3) Bradshaw D, van-Wyk VP, Neethling I, Roomaney RA, Cois A, Joubert JD, et al. Overview: Second Comparative Risk Assessment for South Africa (SACRA2) highlights need for health promotion and strengthened surveillance. *South African Medical Journal*. 2022;556–70.
- 4) van-Wyk VP, Cois A, Kengne AP, Roomaney RA, Levitt N, Turawa EB, et al. Estimating the changing burden of disease attributable to high fasting plasma glucose in South Africa for 2000, 2006 and 2012. *South African Medical Journal*. 2022;594–606.
- 5) Matzopoulos R, Cois A, Probst C, Parry CDH, Vellios N, Sorsdahl K, et al. Estimating the changing burden of disease attributable to alcohol use in South Africa for 2000, 2006 and 2012. *South African Medical Journal*. 2022;662–75.
- 6) English R, Cois A, Tanna G, Day C, Zondi T. Demographic and health trends: 2015-2020. In: *The South African Health Reforms 2015 - 2020: The Road Ahead*. Johannesburg: Trackstar Trading 111 (Pty) Ltd; 2022. ISBN:978-0-639-72368-6
- 7) Awotiwon OF, Cois A, Pacella R, Turawa EB, Dhansay MA, Stuijvenberg L, et al. Estimating the changing disease burden attributable to iron deficiency in South Africa, 2000, 2006 and 2012. *South African Medical Journal*. 2022;684–92.
- 8) Jabar A, Oni T, London L, Cois A, Matzopoulos R. Matching study using health and police datasets for characterising interpersonal violence in the community of Khayelitsha, South Africa 2013–2015. *BMJ Open*. 2022 Sep 1;12(9):e048129.
- 9) Roomaney RA, Wright CY, Cairncross E, Abdelatif N, Cois A, Turawa EB, et al. Estimating the burden of disease attributable to household air pollution from cooking with solid fuels in South Africa for 2000, 2006 and 2012. *South African Medical Journal*. 2022;718–28.
- 10) Roomaney RA, van Wyk B, Cois A, Pillay-van Wyk V. Inequity in the Distribution of Non-Communicable Disease Multimorbidity in Adults in South Africa: An Analysis of Prevalence and Patterns. *Int J Public Health*. 2022;162
- 11) Roomaney RA, Cairncross E, Tesfaye M, Kapwata T, Abdulatif N, Olivier C, Mathibela K, Cois A et al. Estimating the burden of disease attributable to ambient air pollution (ambient PM_{2.5} and ambient ozone) in South Africa for 2000, 2006 and 2012. *South African Medical Journal*. 2022;705–17.
- 12) Nojilana B, Peer N, Abdelatif N, Cois A, Schutte AE, Labadarios D, et al. Estimating the changing burden of disease attributable to high systolic blood pressure in South Africa for 2000, 2006 and 2012. *South African Medical Journal*. 2022;571–82.
- 13) Nojilana B, Abdelatif N, Cois A, Schutte AE, Wentzel-Viljoen E, Turawa EB, et al. Estimating the changing burden of disease attributable to high sodium intake in South Africa for 2000, 2006 and 2012. *South African Medical Journal*. 2022;627–38.
- 14) Neethling I, Peer N, Cois A, Nojilana B, Pacella R, Bradshaw D, et al. Estimating the changing disease burden attributable to raised low-density lipoprotein cholesterol in South Africa for 2000, 2006 and 2012. *South African Medical Journal*. 2022;607–16.

- 15) Neethling I, Lambert EV, Cois A, Roomaney RA, Awotiwon OF, Pacella R, et al. Estimating the changing burden of disease attributable to low levels of physical activity in South Africa for 2000, 2006 and 2012. *South African Medical Journal*. 2022;639–48.
- 16) Nannan N, Neethling I, Cois A, Laubscher R, Turawa EB, Pacella R, et al. Estimating the changing burden of disease attributable to unsafe water and lack of sanitation and hygiene in South Africa for 2000, 2006 and 2012. *South African Medical Journal*. 2022;729–36.
- 17) Bradshaw D, Joubert JD, Abdelatief N, Cois A, Turawa EB, Awotiwon OF, et al. Estimating the changing burden of disease attributable to high body mass index in South Africa for 2000, 2006 and 2012. *South African Medical Journal*. 2022;583–93.
- 18) Kotzé-Hörstmann L, Cois A, Johnson R, et al. Characterization and Comparison of the Divergent Metabolic Consequences of High-Sugar and High-Fat Diets in Male Wistar Rats. *Frontiers in Physiology*. 2022;13.
- 19) van der Hoven J, Allen E, Cois A, et al. Determining antenatal medicine exposures in South African women: a comparison of three methods of ascertainment. *BMC Pregnancy and Childbirth*. 2022;22(1):466. doi:10.1186/s12884-022-04765-1
- 20) Roomaney RA, Wyk B van, Cois A, Wyk VP van. One in five South Africans are multimorbid: An analysis of the 2016 demographic and health survey. *PLOS ONE*. 2022;17(5):e0269081. doi:10.1371/journal.pone.0269081
- 21) Adjaye-Gbewonyo K, Cois A. Explaining population trends in cardiovascular risk: protocol for a comparative analysis of health transitions in South Africa and England using nationally representative survey data. *BMJ Open* 2022;12:e061034.
- 22) Roomaney RA, van Wyk B, Cois A and Pillay-van Wyk V (2022) Multimorbidity Patterns in a National HIV Survey of South African Youth and Adults. *Front. Public Health* 10:862993.

2021

- 23) Hendricks M, Cois A, Geel J, et al. Sex Cord Stromal Tumors in Children and Adolescents: A First Report by The South African Children's Cancer Study Group (1990-2015). *Journal of Pediatric Hematology/Oncology*. 2021;43(5):e619–e624.
- 24) Turawa E, Awotiwon O, Dhansay M A, Cois A, Labadarios D, Bradshaw D, Pillay-van Wyk V. Prevalence of Anaemia, Iron Deficiency, and Iron Deficiency Anaemia in Women of Reproductive Age and Children under 5 Years of Age in South Africa (1997-2021): A Systematic Review. *International Journal of Environmental Research and Public Health*. 2021;18(23)
- 25) Neethling I, Peer N, Cois A, Nojilana B, Pacella R, Bradshaw D, Pillay-van Wyk V. Trends and attributable burden of raised low-density lipoprotein cholesterol in South Africa. *International Journal of Epidemiology*. 2021;50(Supplement_1)
- 26) Day C, Gray A, Cois A, Ndlovu N, Massyn N, Boerma T. Is South Africa closing the health gaps between districts? Monitoring progress towards universal health service coverage with routine facility data. *BMC Health Services Research*. 2021;21(1):194.
- 27) Hendricks M, Cois A, Geel J, van Heerden J, Naidu G, du Plessis J, Bassingthwaighe M, van Zyl A, Uys R, Buchner A, Rowe B, Omar F, Mahlachana N, Thomas K, Vermeulen J, Davidson A, Donald K A, Kruger M. Sex Cord Stromal Tumors in Children and Adolescents: A First Report by The South African Children's Cancer Study Group (1990-2015). *Journal of Pediatric Hematology/Oncology*. 2021;43(5):e619–e624.
- 28) Cois A, Matzopoulos R, Pillay-van Wyk V, Bradshaw D. Bayesian modelling of population trends in alcohol consumption provides empirically based country estimates for South Africa. *Popul Health Metr*. 2021;19(1):43.

2020

- 29) Day C, Gray A, Padayachee T, Cois A. Health and related indicators. In: Kathard H, Padarath A, Galvaan R, Lorenzo T (Eds). *South African Health Review*. Durban: Health Systems Trust; 2020.
- 30) Malaba TR, Cois A, Madlala HP, Matjila M, Myer L, Newell ML, for the PIMS Study Group. Blood pressure trajectories during pregnancy and associations with adverse birth outcomes among HIV-infected and HIV-uninfected women in South Africa: a group-based trajectory modelling approach. *BMC Pregnancy and Childbirth*. 2020;20(1):742.

- 31) Knight D, Ehrlich R, Cois A, Fielding K, Grant A D, Churchyard G. Predictors of silicosis and variation in prevalence across mines among employed gold miners in South Africa. BMC Public Health [Internet]. 2020;20(1):829.
- 32) Cois A, Kengne AP. Non-communicable diseases. In: Massyn N, Barron P, Day C, Ndlovu N, Padarath A (Editors). District Health Barometer 2018/19. Durban: Health Systems Trust; 2020.
- 33) Massyn N, Day C, Ndlovu N, Cois A. Overview of universal health coverage and the health-related Sustainable Development Goals. In: Massyn N, Barron P, Day C, Ndlovu N, Padarath A (Editors). District Health Barometer 2018/19. Durban: Health Systems Trust; 2020.

prior to 2020

- 34) Day C, Gray A, Ndlovu N, Cois A. Health and related indicators: interrogating the UHC service coverage index. In: Moeti T, Padarath A, editors. South African Health Review 2019. Durban: Health Systems Trust; 2019. <http://www.hst.org.za/publications/Pages/SAHR2019>
- 35) Garman EC, Cois A, Schneider M, Lund C. Association between perinatal depressive symptoms and suicidal risk among low-income South African women: a longitudinal study. Social Psychiatry and Psychiatric Epidemiology: 2019.
- 36) Garman EC, Cois A, Tomlinson M, Rotheram-Borus MJ, Lund C. Course of perinatal depressive symptoms among South African women: associations with child outcomes at 18 and 36 months. Social Psychiatry and Psychiatric Epidemiology:2019.
- 37) Cois A, Ehrlich R. Antihypertensive treatment and blood pressure trends among South African adults: A repeated cross-sectional analysis of a population panel survey. PLOS ONE 13(8): e0200606, 2018.
- 38) Lund C, Cois A. Simultaneous social causation and social drift: Longitudinal analysis of depression and poverty in South Africa. J Affect Disord 229:396–402, 2018.
- 39) Kanyik JPM, Cois A, Kalla AA. *The spectrum of psoriatic arthritis in a South African cohort*. Clin. Rheumatol. 2017 36(11):2501–2507, 2017.
- 40) Cois A. *Understanding blood pressure dynamics in the South African population: a latent variables approach to the analysis and comparison of data from multiple surveys*. PhD Thesis. University of Cape Town, Faculty of Health Sciences. 2017. Available at: <http://hdl.handle.net/11427/25196>
- 41) Cois, A. *Non communicable diseases (book chapter)*. In N. Massyn, N. Peer, R. English, A. Padarath, P. Barron, & C. Day (Eds.), District Health Barometer 2015/2016. Durban: Health Systems Trust, 2016
- 42) Phillips T, Cois A, Remien RH, Mellins CA, McIntyre JA, Petro G, Abrams EJ, Myer L. *Self-reported Side Effects and Adherence to Antiretroviral Therapy in HIV-infected Pregnant Women under Option B+: a prospective study*. PLOS ONE 11(10): e0163079, 2016.
- 43) Cois A, Ehrlich R. *Socioeconomic Status Modifies the Seasonal Effect on Blood Pressure: Findings From a National Panel Study*. Medicine, 94(35), 2015.
- 44) Cois A, Day C. *Obesity trends and risk factors in the South African adult population*. BMC Obesity, 2(1), 2015.
- 45) Cois, A, Day, C. Ranking district health system performance (book chapter). In N. Massyn, C. Day, N. Peer, A. Padarath, P. Barron, & R. English (Eds.), District Health Barometer 2013/2014 (pp. 316–338). Durban: Health Systems Trust, 2015.
- 46) Cois A, Ehrlich R. *Analysing the socioeconomic determinants of hypertension in South Africa: a structural equation modelling approach*. BMC Public Health, 14(1), 2014.
- 47) Cois A. *Dealing with multiple confounders in observational studies: beyond logistic regression*. Sacema Quarterly, March, 2014.
- 48) Cois A, Ehrlich R. *Problem drinking as a risk factor for tuberculosis: a propensity score matched analysis of a national survey*. BMC Public Health, 13, 871, 2013.
- 49) Lorenzo T, Ned-Matiwane L, Cois A, Nwanze I. *Youth, Disability and rural communities: facing the challenges of change*. Disability Catalyst Africa Series no. 3. Cape Town: Disability Innovation Africa & University of Cape Town, 2013.
- 50) Cois A, [Social determinants and addictions: from observation to intervention] In Italian. Medicina delle Dipendenze – The Italian Journal of the Addictions, 11(5), 2012.
(Title translation:)

- 51) Cois A, *Dentro la scatola nera: processi ed elementi attivi nei trattamenti residenziali*, Medicina delle Tossicodipendenze – The Italian Journal of the addiction, XVII (65), 2009.
(Title translation: *Inside the black box: processes and active elements in residential treatments*)
- 52) Cois A, [Dual Diagnosis] *In Italian*. Appunti Sparsi, ANNO XVI, Numero 3, Carbonia (CA), 2006, 22-28.
- 53) Vidotto F, Cois A, Cester A, Morittu S, Vivaldelli A, [Methodology and tools for multidimensional evaluation: an introduction] *In Italian. (book chapter)*, in: AA.VV., Integrazione Comunicazione nel settore socio-sanitario: metodi e strumenti, Mareno di Piave: Edizioni Vega, 2004, 259-279. ISBN: 8886933339.
- 54) Cois A, Lai Guaita MP. [Cannabis and Ecstasy: everybody's experience?] *In Italian. (book chapter)*, in Lai Guaita M.P., L'Ecstasy e le altre droghe, Carocci, Roma, 1999, 179-189. ISBN: 8843012614.
- 55) Cois A, Morittu S. [Drug addicts in the therapeutic community and their families: two parallel journeys?] *In Italian. (Book Chapter)*, in Lai Guaita M.P, Famiglia e Tossicodipendenza, Laterza, 1996. ISBN: 8842051071.
- 56) Cois A, Ganadu L, Morittu S. [Follow-up study of 545 opioid addicts admitted to a residential community between 1980 and 1992] *In Italian*. Uno Studio di Follow-Up Su 545 Eroinomani Ammessi in Comunità Residenziale tra il 1980 e il 1992, Bollett per le farmac e l'alcool, XVIII(3), 1995.
- 57) Cois A, Lamieri P, Marongiu I. A variable reluctance motor with optimized rotor anisotropy. Proc. of the International Conference on the Evolution and Modern Aspects of Synchronous Machines, Part 2. Zurich, Switzerland, 1991.
- 58) Cois A, Marongiu I. A variable reluctance motor drive for electrical vehicle applications. 24th ISATA International Symposium on automotive technology and automation, Florence, Italy, 1991, 111-118
- 59) Cois A, Fanni A, Giua A. An expert system for designing and supervising a discrete event model. 1990. Proc. of the IEEE International Workshop on Intelligent Motion Control, Vol. 1, Istanbul, Turkey, 1990. 103-107
- 60) Cois A, Fanni A, Pes S, Giua A. Qualitative analysis in simulating analog circuits. Proc. of the IEEE International Symposium on Intelligent Control, Albany, NY, 1989. 531-536

Conference Presentations

- 1) Cois A. *Bayesian meta-regression models for the estimation of population trends in health risk factors*. South African Statistical Association Conference. George, 30 November -1 December 2022.
- 2) Cois A, Nkonki L. *SurveyLab: a virtual environment for teaching and learning survey methods*. Oral Presentation. Stellenbosch University SOTL Conference 2022. October 2022.
- 3) Cois A, Matzopoulos R. *Estimating age-and sex-specific trends in alcohol use in South Africa: a Bayesian approach*. Stellenbosch University Academic Day (virtual). 18 August 2022.
- 4) Cois A, Padayachee T, Andrews G. *AI for real-time monitoring of health research trends in South Africa*. Sixth Global Symposium on Health Systems Research. Dubai 8-12 November 2020 (remote presentation).
- 5) Cois A, Padayachee T, Muthivhi T, Malinga L, Zondi T. *Trends in research conducted in public health facilities in South Africa 2015-2018: an analysis of the National Health Research Database*. PHASA Annual Conference 2019. Cape Town 16-18 September 2019.
- 6) Cois A. *Who drinks in South Africa, and how much? Integrating multiple sources of information to estimate alcohol consumption trends*. StanCon 2018. Helsinki, Finland 29/31 August 2018.
- 7) Cois A, Solomons W, English R. *Improving patient experience in primary health care: a simulation study on the effect of organisational changes on waiting times*. PHASA Conference. Johannesburg 4-7 September 2017.
- 8) Cois, A. When shape matters: trends in general and abdominal obesity in the South African Adult population, 1998-2015. Invited Talk. Africa Health Public Health Conference. Johannesburg 7-8 June 2017.
- 9) Cois A, *Brief interventions in preventing drug and alcohol abuse in rural areas: a pilot study*, 14th Annual Conference on Rural Health, 26-28 August 2010 - Ezulwini Valley, Swaziland.

- 10) Cois A, Congia P, [*Parenting, attachment style and personality disorders in a sample of drug dependent individuals*] (*In Italian*), Convegno Nazionale "Fragil... mente: Tossicodipendenza e malattia mentale dall'adolescenza all'età adulta", Forte Village Resort (CA), 4 – 8 October 2004.
- 11) Cois A, Morittu S, [*Personality structure and residential treatment compliance: which relationships?*] *in Italian*. Convegno Nazionale "Aggiornamenti nella terapia delle tossicodipendenze, Cagliari, 29-31 January 2004.
- 12) Cois A, Morittu S, [Psychiatric comorbidity in a high-threshold residential treatment: the experience of the Associazione Mondo X – Sardegna] *In Italian*. Convegno Nazionale Aggiornamenti nella terapia delle tossicodipendenze, Cagliari, 29-31 January, 2004.
- 13) Cois A, Morittu S, Murgia S, Portas M, [The impact of psychiatric comorbidity in a residential service for drug addicts: personality and psychopathologic characteristics of the patients]. Proc. 1° Congresso Nazionale su Disturbi Mentali e Tossicodipendenza, Chia Laguna, Domus de Maria, 2002.
- 14) Cois A, Congia P, Portas M, [Dual diagnosis and therapeutic community treatments: remarks on retention predictors] *In Italian*. 1° congresso Nazionale su disturbi mentali e tossicodipendenza, Chia Laguna, Domus De Maria (CA), 2002.

Languages and personal skills

Mother Tongue Italian

Other Languages English (Fluent)

French (Moderate written, basic spoken)

Other skills Knowledge of: Microsoft® Windows® Operating Systems, Office suite, *LaTeX* writing environment; programming/scripting languages (javascript, vb.net/c#). Advanced knowledge of software packages for statistical computation and data management (R™, Stata®, Mplus®, Stan™)

Cape Town, 1st of January 2023.